

Thermodynamics Past Exam Booklet

Weeks 1–6 Questions and Clear Answers

Mid-semester papers plus selected end-of-year applications

How to use this booklet. Attempt each question before reading its answer. Every solution identifies the system, assumptions, property model, sequential working, units, final result and common trap. Repeated questions are retained because repetition reveals what examiners value. End-of-year questions are clearly marked as stretch applications.

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1 Coverage map and source boundary

This booklet contains **22 rewritten past-exam questions**: nineteen from five mid-semester papers and three selected end-of-year refrigeration questions that directly extend Week 5–6 principles.

Source	Questions	Main coverage
Test 1 (2016)	4	phase diagrams, R-134a piston-cylinder, rigid tank, water tables
Test 2 (2018)	4	ideal-gas validity, R-134a tables, piston stops, electrical/paddle work
Test 3 (2019)	4	phase/table skills, electrical heating, ideal-gas process cycle
Test 4 (2017 alt.)	4	repeated phase skill, steam piston, room heating, R-134a tables
2025 test	3	sensible heat, spring-loaded piston, condenser heat exchanger
End-year selection	3	Week 5/6 refrigeration balances and COP as stretch applications

Evidence rule. Official handwritten answers are followed where internally consistent. If a source answer is ambiguous or arithmetically inconsistent, this booklet says so and gives a transparent reconstruction. Property values remain tied to the edition of the course tables used by the paper, so small last-digit differences may occur with another table edition.

What was excluded from the end-of-year set. Otto, Diesel, detailed Brayton-regenerator and Rankine-cycle problems were not included because they depend on later cycle theory rather than only the first six weeks. The selected refrigeration questions use Week 5 steady-flow balances and Week 6 COP concepts, but are labelled as stretch work because they also require vapour-compression-cycle conventions.

Test 1

Q1

Rewritten question. For the unlabeled P – v diagram of a pure substance, identify the phase regions, key features, and process lines requested in Figure 1.

Source page/marks. source/test1.pdf, p. 2; 5 marks.

Vague-data note. This is a diagram-labelling question, so the answer is taken from the handwritten labelled solution page rather than from a numerical calculation.

System and boundary. Not a calculation problem; a conceptual pure-substance phase map on a P – v diagram.

Assumptions.

- The sketch is the standard pure-substance P – v phase diagram.
- The handwritten labels on the solution page are the official intended identifications.
- The constant-temperature boiling path runs from compressed liquid through the two-phase dome to superheated vapour.

Given / Find.

- Given: an unlabeled P – v diagram for a pure substance.
- Find: liquid, solid, vapour, liquid+solid, solid+vapour, liquid+vapour, critical point, triple line, constant-pressure sublimation path, and constant-temperature boiling path.

Governing equations. Qualitative phase-diagram topology only; no numerical equations are required.

Sequential working.

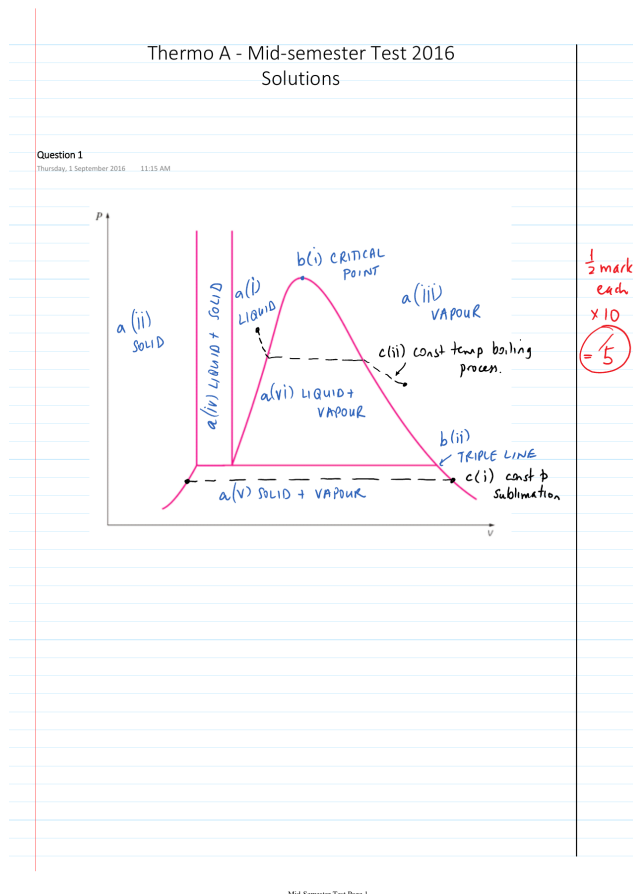
- Left region: solid.
- Between the left boundary and the left side of the triple region: liquid+solid.
- Inside the dome: liquid+vapour.
- Right of the dome: vapour.
- Below the triple-point horizontal segment: solid+vapour.
- The apex of the dome: critical point.
- The horizontal coexistence line at low pressure: triple line.
- Constant-pressure sublimation follows the solid–vapour boundary.
- Constant-temperature boiling follows the line from compressed liquid to superheated vapour through the two-phase region.

Property-table values. None.

Boxed final answer.

Six phase regions + critical point + triple line
sublimation path + boiling path

Exam trap. Do not place the triple line on the dome boundary; it is the low-pressure horizontal coexistence line.

Figure 1: Official Q1 P - v diagram with labels.

Q2

Rewritten question. A piston-cylinder device contains 3 kg of R-134a at 200 kPa and quality $x = 0.2$. Heat is added until the pressure reaches 400 kPa, after which the piston moves at constant pressure until the volume becomes twice the initial volume. Determine the initial volume, the boundary work, and the total heat transfer.

Source page/marks. source/test1.pdf, pp. 3–4; 16 marks.

Vague-data note. The source diagram is only partially informative, so the process description is taken from the printed text and verified against the handwritten solution: constant-volume heating to 400 kPa, then constant-pressure expansion to $V_3 = 2V_1$.

System and boundary. Closed system; the refrigerant is the control mass inside the piston-cylinder device. The boundary is the moving piston face and cylinder walls.

Assumptions.

- R-134a properties are read from the saturation tables used in the handwritten solution.
- The process from 1 to 2 occurs at constant volume while the piston is on the stops, so $W_{1 \rightarrow 2} = 0$.
- The process from 2 to 3 occurs at constant pressure 400 kPa.

Given / Find.

- Given: $m = 3$ kg, $P_1 = 200$ kPa, $x_1 = 0.2$, $P_3 = 400$ kPa, $V_3 = 2V_1$.
- Find: V_1 , $W_{1 \rightarrow 3}$, and $Q_{1 \rightarrow 3}$.

Governing equations.

$$\begin{aligned} V_1 &= mv_1, & v_1 &= v_f + x_1 v_{fg} \\ W_{1 \rightarrow 2} &= 0, & W_{2 \rightarrow 3} &= P_3 (V_3 - V_2) \\ Q_{1 \rightarrow 3} - W_{1 \rightarrow 3} &= \Delta U = m(u_3 - u_1) \end{aligned}$$

Property-table values.

- At 200 kPa: $v_f = 0.0007532$ m³/kg, $v_g = 0.09951$ m³/kg, $u_f = 38.26$ kJ/kg, $u_{fg} = 186.25$ kJ/kg.
- At 400 kPa: $v_f = 0.0007905$ m³/kg, $v_g = 0.051266$ m³/kg, $u_f = 63.61$ kJ/kg, $u_{fg} = 171.49$ kJ/kg.

Sequential working.

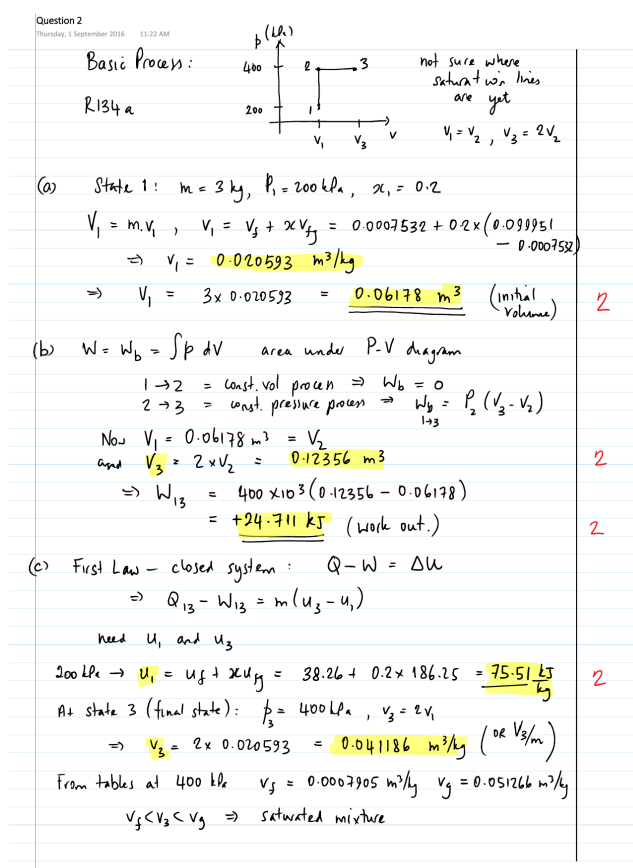
$$v_1 = 0.0007532 + 0.2(0.09951 - 0.0007532) = 0.020593 \text{ m}^3/\text{kg}$$

$$\begin{aligned}
 V_1 &= 3(0.020593) = 0.06178 \text{ m}^3 \\
 V_2 &= V_1 = 0.06178 \text{ m}^3, \quad V_3 = 2V_1 = 0.12356 \text{ m}^3 \\
 W_{1 \rightarrow 3} &= W_{2 \rightarrow 3} = 400 \times 10^3 (0.12356 - 0.06178) = 24.711 \text{ kJ} \\
 u_1 &= 38.26 + 0.2(186.25) = 75.51 \text{ kJ/kg} \\
 v_3 &= \frac{V_3}{m} = \frac{0.12356}{3} = 0.041186 \text{ m}^3/\text{kg} \\
 x_3 &= \frac{v_3 - v_f}{v_{fg}} = \frac{0.041186 - 0.0007905}{0.051266 - 0.0007905} = 0.800 \\
 u_3 &= 63.61 + 0.8(171.49) = 200.8 \text{ kJ/kg} \\
 Q_{1 \rightarrow 3} &= W_{1 \rightarrow 3} + m(u_3 - u_1) = 24.711 + 3(200.802 - 75.51) = 400.6 \text{ kJ}
 \end{aligned}$$

Boxed final answer.

$$V_1 = 0.06178 \text{ m}^3, \quad W_{1 \rightarrow 3} = 24.711 \text{ kJ}, \quad Q_{1 \rightarrow 3} = 400.6 \text{ kJ}$$

Exam trap. The first leg is constant volume, so its boundary work is zero even though heat is added.



Mid-Semester Test Page 2

Figure 2: Official Q2 piston-cylinder apparatus and piecewise P - v process working.

Q3

Rewritten question. A rigid tank is divided into two equal parts by a partition. Initially one side contains 5 kg of water at 200 kPa and 25°C, and the other side is evacuated. The partition is removed and the water expands to fill the entire tank, then exchanges heat with the surroundings until the temperature returns to 25°C. Determine the tank volume, the final pressure, and the heat transfer.

Source page/marks. source/test1.pdf, p. 5; 16 marks.

Vague-data note. The handwritten solution uses the 25°C saturated-water properties and treats the initial state as compressed liquid; the tank is rigid and the final temperature is forced back to 25°C.

System and boundary. Closed system; the water in the rigid tank is the control mass. The boundary is fixed, so there is no boundary motion.

Assumptions.

- Water at 25°C and 200 kPa is a compressed liquid, so $v_1 \approx v_f@25^\circ\text{C}$.
- The tank is rigid, so the total volume is constant and $W_b = 0$.
- Final state is at 25°C and the same specific volume set by the expanded rigid tank.

Given / Find.

- Given: $m = 5 \text{ kg}$, $P_1 = 200 \text{ kPa}$, $T_1 = 25^\circ\text{C}$, initially one half evacuated.
- Find: V_{tank} , P_2 , and $Q_{1 \rightarrow 2}$.

Governing equations.

$$v_1 \approx v_f@25^\circ\text{C}, \quad V_1 = mv_1, \quad V_{\text{tank}} = 2V_1$$

$$v_2 = \frac{V_{\text{tank}}}{m}, \quad Q_{1 \rightarrow 2} = \Delta U = m(u_2 - u_1)$$

Property-table values.

- At 25°C: $v_f = 0.001003 \text{ m}^3/\text{kg}$, $v_g = 43.34 \text{ m}^3/\text{kg}$, $u_f = 104.83 \text{ kJ/kg}$, $u_{fg} = 2304.3 \text{ kJ/kg}$.
- Saturation pressure at 25°C: $P_{\text{sat}} = 3.1698 \text{ kPa}$.

Sequential working.

$$v_1 \approx 0.001003 \text{ m}^3/\text{kg}$$

$$V_1 = 5(0.001003) = 0.005015 \text{ m}^3$$

$$V_{\text{tank}} = 2V_1 = 0.01003 \text{ m}^3$$

$$v_2 = \frac{0.01003}{5} = 0.002006 \text{ m}^3/\text{kg}$$

$$x_2 = \frac{v_2 - v_f}{v_{fg}} = \frac{0.002006 - 0.001003}{43.34 - 0.001003} = 2.314 \times 10^{-5}$$

$$u_1 = 104.83 \text{ kJ/kg}$$

$$u_2 = u_f + x_2 u_{fg} = 104.83 + (2.314 \times 10^{-5})(2304.3) = 104.88 \text{ kJ/kg}$$

$$P_2 = P_{\text{sat}}@25^\circ\text{C} = 3.1698 \text{ kPa}$$

$$Q_{1 \rightarrow 2} = 5(104.88 - 104.83) = 0.267 \text{ kJ}$$

Boxed final answer.

$V_{\text{tank}} = 0.01003 \text{ m}^3, \quad P_2 = 3.1698 \text{ kPa}, \quad Q_{1 \rightarrow 2} = 0.267 \text{ kJ}$

Exam trap. The tank is rigid, so the pressure change comes from heat transfer and phase change, not from boundary work.

- The volume of the tank.
- The final pressure.
- The heat transfer for this process.

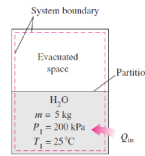


Figure 3

[illegible]

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Figure 3: Rigid-tank diagram used in Q3.

Q4

Source page/marks. source/test1.pdf, p. 7; 9 marks.

System and boundary. Property-table lookup only; no control volume or control mass boundary is needed.

- Steam tables for water are used.
- $x = 0$ means saturated liquid, $x = 1$ means saturated vapour, and an omitted x means it is not applicable.
- When u lies between u_f and u_g , the state is a saturated mixture and $u = u_f + xu_{fg}$.

- Given: the partial table entries for rows a–f.
- Find: every missing T , p , u , x , and phase description entry.

$$\begin{aligned} x &= \frac{u - u_f}{u_{fg}}, & u &= u_f + xu_{fg} \\ P &= P_{sat}(T), & T &= T_{sat}(P) \end{aligned}$$

- Row (a), 145°C: $P_{sat} = 415.68 \text{ kPa}$, $u_f = 610.19 \text{ kJ/kg}$, $u_{fg} = 1944.2 \text{ kJ/kg}$.
- Row (b), 800 kPa: $T_{sat} = 170.41^\circ\text{C}$, but $u = 209.33 \text{ kJ/kg} < u_f$, so the state is compressed liquid; the handwritten completion gives $T = 50^\circ\text{C}$.
- Row (c), 270°C: $P_{sat} = 5503.0 \text{ kPa}$ and $u_g = 2593.7 \text{ kJ/kg}$.
- Row (d), 30 kPa: $T_{sat} = 69.09^\circ\text{C}$, $u_f = 289.24 \text{ kJ/kg}$, $u_{fg} = 2178.5 \text{ kJ/kg}$.
- Row (e), 1600 kPa and $u = 3120.1 \text{ kJ/kg}$: superheated vapour; handwritten completion gives $T = 500^\circ\text{C}$.
- Row (f), 85°C: $P_{sat} = 57.868 \text{ kPa}$, $u_f = 355.96 \text{ kJ/kg}$.

Sequential working.

- (a) $x = (2000 - 610.19)/1944.2 = 0.7148$; $P = 415.68$ kPa; saturated mixture.
- (b) At 800 kPa, $u < u_f$, so the state is compressed liquid; the completed table gives $T = 50^\circ\text{C}$.
- (c) Saturated vapour at 270°C gives $x = 1$ and $u = 2593.7$ kJ/kg.
- (d) $x = 0.4$ gives $u = 289.24 + 0.4(2178.5) = 1160.64$ kJ/kg and $T = 69.09^\circ\text{C}$.
- (e) Interpolating the superheated tables at 1.6 MPa gives $u = 3120.1$ kJ/kg at $T = 500^\circ\text{C}$.
- (f) $x = 0$ gives saturated liquid at 85°C with $P = 57.868$ kPa and $u = 355.96$ kJ/kg.

Completed final table.

Row	$T(^{\circ}\text{C})$	$p(\text{kPa})$	$u(\text{kJ/kg})$	x	Phase
a	145	415.68	2000	0.7148	saturated mixture
b	50	800	209.33	–	compressed liquid
c	270	5503.0	2593.7	1.0	saturated vapour
d	69.09	30	1160.64	0.4	saturated mixture
e	500	1600	3120.1	–	superheated vapour
f	85	57.868	355.96	0.0	saturated liquid

All six states classified before table lookup.
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Exam trap. For row (b), do not force a quality value into a compressed-liquid state; x is only defined on the saturation line.

Test 2

Q1

Rewritten question. Under what conditions (temperature, pressure, density) could the ideal gas equation be used to accurately predict properties for steam?

Vague/source note. The source page gives only the qualitative prompt; the handwritten response is a three-point list and no numerical values are required.

System and boundary. Closed-system property statement for steam; no process calculation is needed.

Assumptions.

- Steam is sufficiently far from the saturation dome that ideal-gas behavior is a good approximation.
- The pressure is low.
- The density is low, which corresponds to a large specific volume.

Given / Find.

- Given: steam, with the condition asked qualitatively in terms of temperature, pressure, and density.
- Find: the conditions under which the ideal-gas equation is accurate.

Governing equations.

$$PV = mRT, \quad Z \approx 1 \text{ for ideal-gas behavior}$$

Property-table values. None.

Sequential working with units.

- High temperature.
- Low pressure.
- Low density.

Boxed official answer.

Use the ideal-gas equation for steam at high temperature, low pressure, and low density.

Exam trap. Do not say only “high temperature”; the handwritten answer explicitly includes low pressure and low density as well.

Source pages / marks. Test sheet p. 2 / 3 marks.

Q2

Rewritten question. Fill in the missing information in the following table for R134a.

Vague/source note. The source page shows a filled table and two worked parts below it. The handwritten values in the table are the official results to preserve.

System and boundary. Property-state lookup for R134a; each row is a separate equilibrium state.

Assumptions.

- Each table entry is read from the appropriate saturated, compressed-liquid, or superheated R134a data.
- Saturated-mixture rows use the quality relation $h = h_f + xh_{fg}$.
- The handwritten values on the source page are taken as the target answers.

Given / Find.

- Given: the partially completed R134a table below.
- Find: the missing T , p , h , x , and phase description entries.

Governing equations.

$$h = h_f + xh_{fg}, \quad x = \frac{h - h_f}{h_{fg}} \quad \text{for saturated mixtures}$$

Property-table values.

	T (°C)	p (kPa)	h (kJ/kg)	x / phase description
a.	−22	121.72	22.89	0; saturated liquid
b.	31.31	800	200	0.6; saturated mixture
c.	−20	132.82	89.36	0.3; saturated mixture
d.	8	1000	62.69	−; compressed liquid
e.	40	200	287.74	−; superheated
f.	85	2928.2	279.51	1.0; saturated vapour

Sequential working with units.

$$\text{b) } x = \frac{h - h_f}{h_{fg}} = \frac{200 - 95.48}{174.06} = 0.6$$

$$\text{c) } h = h_f + xh_{fg} = 25.47 + 0.3 \times 212.96 = 89.36 \text{ kJ/kg}$$

Boxed official answer.

- a. $T = -22^\circ\text{C}$, $p = 121.72 \text{ kPa}$, $h = 22.89 \text{ kJ/kg}$, $x = 0$, saturated liquid
 b. $T = 31.31^\circ\text{C}$, $p = 800 \text{ kPa}$, $h = 200 \text{ kJ/kg}$, $x = 0.6$, saturated mixture
 c. $T = -20^\circ\text{C}$, $p = 132.82 \text{ kPa}$, $h = 89.36 \text{ kJ/kg}$, $x = 0.3$, saturated mixture
 d. $T = 8^\circ\text{C}$, $p = 1000 \text{ kPa}$, $h = 62.69 \text{ kJ/kg}$, $x = --$, compressed liquid
 e. $T = 40^\circ\text{C}$, $p = 200 \text{ kPa}$, $h = 287.74 \text{ kJ/kg}$, $x = --$, superheated
 f. $T = 85^\circ\text{C}$, $p = 2928.2 \text{ kPa}$, $h = 279.51 \text{ kJ/kg}$, $x = 1.0$, saturated vapour

Exam trap. The source has mixed state types: not every row uses quality; rows d and e are not saturated-mixture states.

Source pages / marks. Test sheet p. 3 / 9 marks.

Q3

Rewritten question. A piston-cylinder device (Figure 1) with a set of stops on the top contains 2.2 kg of saturated liquid water at 150 kPa. Heat is transferred to the water, causing some of the liquid to evaporate and move the piston upwards (at constant pressure). When the piston reaches the stops, the enclosed volume is 45 l. More heat is transferred until the pressure is doubled. Determine: (a) the amount of saturated liquid, if any, in kg, at the final state; (b) the final temperature; (c) the total work and heat transfer; and (d) show the process on a p - v diagram (with respect to saturation lines).

Vague/source note. The handwritten solution uses saturated-water tables at 150 kPa and 300 kPa, then evaluates the final state as a saturated mixture at 300 kPa.

System and boundary. Closed system; moving-boundary piston-cylinder with stops, so boundary work occurs only during the constant-pressure part up to the stops.

Assumptions.

- Saturated-water properties are taken from the steam tables.
- The piston moves at constant pressure until it hits the stops; after that the volume is fixed.
- Kinetic and potential energy changes are negligible.

Given / Find.

- Given: $m = 2.2 \text{ kg}$, initial saturated liquid water at $p_1 = 150 \text{ kPa}$, stop volume $V_2 = 45 \text{ l} = 0.045 \text{ m}^3$, and final pressure $p_3 = 300 \text{ kPa}$.
- Find: $m_{f,3}$, T_3 , $W_{1 \rightarrow 3}$, and $Q_{1 \rightarrow 3}$.

Governing equations.

$$V = mv, \quad x = \frac{v - v_f}{v_{fg}}, \quad u = u_f + xu_{fg}$$

$$W_{1 \rightarrow 3} = W_{1 \rightarrow 2} = p_1(V_2 - V_1), \quad Q_{1 \rightarrow 3} = W_{1 \rightarrow 3} + m(u_3 - u_1)$$

Property-table values.

$$\text{At } 150 \text{ kPa: } u_1 = u_f = 466.97 \text{ kJ/kg}, \quad v_1 = v_f = 0.001053 \text{ m}^3/\text{kg},$$

$$\text{At } 300 \text{ kPa: } u_f = 561.11, \quad u_{fg} = 1982.1 \text{ kJ/kg},$$

$$v_f = 0.001073, \quad v_{fg} = 0.60582 \text{ m}^3/\text{kg}, \quad T_{sat} = 133.53^\circ\text{C}.$$

Sequential working with units.

$$V_1 = mv_1 = (2.2 \text{ kg})(0.001053 \text{ m}^3/\text{kg}) = 0.0023166 \text{ m}^3$$

$$V_2 = 45 \text{ l} = 0.045 \text{ m}^3, \quad p_2 = p_1 = 150 \text{ kPa}$$

$$V_3 = V_2 = 0.045 \text{ m}^3, \quad v_3 = \frac{V_3}{m} = 0.02045 \text{ m}^3/\text{kg}, \quad p_3 = 2p_1 = 300 \text{ kPa}$$

$$v_f < v_3 < v_g \text{ at } 300 \text{ kPa} \Rightarrow x_3 = \frac{v_3 - v_f}{v_{fg}} = \frac{0.02045 - 0.001073}{0.60582 - 0.001073} = 0.032$$

$$u_3 = u_f + x_3u_{fg} = 561.11 + 0.032 \times 1982.1 = 624.54 \text{ kJ/kg}$$

$$m_{f,3} = (1 - x_3)m = (1 - 0.032)(2.2 \text{ kg}) = 2.12 \text{ kg}$$

$$T_3 = T_{\text{sat}}(300 \text{ kPa}) = 133.53^\circ\text{C}$$

$$W_{1 \rightarrow 3} = W_{1 \rightarrow 2} = p_1(V_2 - V_1) = (150 \text{ kPa})(0.045 - 0.0023166 \text{ m}^3) = 6.4025 \text{ kJ}$$

$$Q_{1 \rightarrow 3} = W_{1 \rightarrow 3} + m(u_3 - u_1) = 6.4025 \text{ kJ} + (2.2 \text{ kg})(624.54 - 466.97) \text{ kJ/kg} = 353.1 \text{ kJ}$$

Boxed official answer.

$$m_{f,3} = 2.12 \text{ kg}, \quad T_3 = 133.53^\circ\text{C}, \quad W_{1 \rightarrow 3} = 6.4025 \text{ kJ}, \quad Q_{1 \rightarrow 3} = 353.1 \text{ kJ}$$

Exam trap. The work stops once the piston hits the stops; the second heating stage changes internal energy only because the volume is fixed.

Source pages / marks. Test sheet pp. 4–5 / 16 marks.

Question 3 - (16 marks)

A piston-cylinder device (Figure 1) with a set of stops on the top contains 2.2 kg of saturated liquid water at 150 kPa. Heat is transferred to the water, causing some of the liquid to evaporate and move the piston upwards (at constant pressure). When the piston reaches the stops, the enclosed volume is 45 l. More heat is transferred until the pressure is doubled. Determine:

- The amount of saturated liquid (if any), in kg, at the final state.
- The final temperature.
- The total work and heat transfer.
- Show the process on a p - v diagram (with respect to saturation lines).

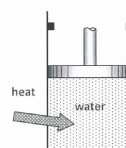


Figure 1

Initial conditions: $p_1 = 150 \text{ kPa}$
 (State 1) $m = 2.2 \text{ kg}$
 $x = 0$
 $u_1 = u_f = 466.97 \text{ kJ/kg}$ ①
 $v_1 = v_f = 0.001053 \text{ m}^3/\text{kg}$ ①
 $\Rightarrow V_1 = m v_1 = (2.2 \text{ kg})(0.001053 \text{ m}^3/\text{kg}) = 0.0023166 \text{ m}^3$ ①
 State 2: $V_2 = 45 \text{ L} = 0.045 \text{ m}^3$
 $p_2 = 150 \text{ kPa}$
 State 3: $V_3 = V_2 = 0.045 \text{ m}^3$
 $\Rightarrow v_3 = V_3/m = 0.02045 \text{ m}^3/\text{kg}$ ①
 $p_3 = 2 \times p_1 = 300 \text{ kPa}$
 $v_f < v_3 < v_g @ 300 \text{ kPa} \Rightarrow \text{saturated mixture}$
 $\Rightarrow x_3 = \frac{v_3 - v_f}{v_{fg}} = \frac{0.02045 - 0.001073}{0.60582 - 0.001073} = 0.032$ ①

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Figure 4: Piston-cylinder device with stops for Q3.

Q4

Rewritten question. An insulated rigid vessel with a volume of 3 m^3 is fitted with a resistance heater and a paddle wheel as shown in Figure 2. The vessel is initially filled with air at a temperature of 27°C and a pressure of 130 kPa . Now, a current of 4 A from a 230 V supply is passed through the resistance heater while the paddle wheel transfers energy at a rate of 150 J/s to the air. Determine the time taken for the air to reach a temperature of 227°C . State any assumptions made and also state any tables used for data.

Vague/source note. The source page and the handwritten solution use the ideal-gas model for air plus temperature-dependent c_v data from Table A-2; a constant- c_v check using Table A-18 is also shown.

System and boundary. Closed rigid vessel; no boundary work, with electrical and shaft work input to the air.

Assumptions.

- Air behaves as an ideal gas in the temperature range considered.
- The vessel is rigid and insulated, so $Q = 0$ and $W_b = 0$.
- Changes in kinetic and potential energy are negligible.

Given / Find.

- Given: $P_1 = 130 \text{ kPa}$, $T_1 = 27^\circ\text{C} = 300 \text{ K}$, $V = 3 \text{ m}^3$, $T_2 = 227^\circ\text{C} = 500 \text{ K}$, $I = 4 \text{ A}$, $V_{el} = 230 \text{ V}$, and paddle-wheel transfer rate 150 J/s to the air.
- Find: time Δt .

Governing equations.

$$Q - W = \Delta U, \quad \Delta U = m(u_2 - u_1), \quad m = \frac{PV}{RT}$$

$$\dot{W}_{el} = VI, \quad \dot{W}_{sh} = 150 \text{ J/s}, \quad \Delta t = \frac{m(u_2 - u_1)}{\dot{W}_{el} + \dot{W}_{sh}}$$

Property-table values.

$$R_{air} = 0.287 \text{ kJ/(kg K)}, \quad c_{v,air} = 0.726 \text{ kJ/(kg K)} \text{ (Table A-2)}$$

Alternative table values shown on the source page: $u_1 = 214.07 \text{ kJ/kg}$ at 300 K , $u_2 = 359.49 \text{ kJ/kg}$ at 500 K

Sequential working with units.

$$m = \frac{PV}{RT_1} = \frac{(130 \text{ kPa})(3 \text{ m}^3)}{(0.287 \text{ kJ/(kg K)})(300 \text{ K})} = 4.53 \text{ kg}$$

$$\Delta u = c_{v,air} \Delta T = (0.726 \text{ kJ/(kg K)})(500 - 300 \text{ K}) = 145.2 \text{ kJ/kg}$$

$$\dot{W}_{el} = -(4 \text{ A})(230 \text{ V}) = -920 \text{ J/s}, \quad \dot{W}_{sh} = -150 \text{ J/s}$$

$$\Delta t = \frac{m(u_2 - u_1)}{-\dot{W}_{el} - \dot{W}_{sh}} = \frac{(4.53 \text{ kg})(145.2 \text{ kJ/kg})}{(920 + 150) \times 10^{-3} \text{ kJ/s}} = 614.7 \text{ s}$$

$$614.7 \text{ s} = 10 \text{ min } 14 \text{ s}$$

Boxed official answer.

$$\Delta t = 614.7 \text{ s} \approx 10 \text{ min } 14 \text{ s}$$

Exam trap. The heater power and paddle-wheel power both enter as energy to the air; do not forget the sign convention when writing the closed-system energy balance.

Source pages / marks. Test sheet pp. 8–9 / 8 marks.

Thermodynamics A – TEST – August 2018

Question 4 - (8 marks)

An insulated rigid vessel with a volume of 3 m^3 is fitted with a resistance heater and a paddle wheel as shown in Figure 2. The vessel is initially filled with air at a temperature of 27°C and a pressure of 130 kPa . Now, a current of 4 A from a 230 V supply is passed through the resistance heater while the paddle wheel transfers energy at a rate of 150 J/s to the air. Determine the time taken for the air to reach a temperature of 227°C .

(State any assumptions made and also state any tables used for data).

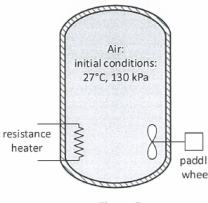


Figure 2

Initial state: $P_1 = 130 \text{ kPa}$
 $T_1 = 27^\circ\text{C} = 300 \text{ K}$
 $V_1 = 3 \text{ m}^3$

Final state: $P_2 = ?$
 $T_2 = 227^\circ\text{C} = 500 \text{ K}$
 $V_2 = V_1$

$Q = 0$
 $-W = \Delta U$

$(-\dot{W}_{el} - \dot{W}_{sh}) \Delta t = m(u_2 - u_1) \quad (2)$

$\dot{W}_{el} = -(4 \text{ A} \times 230 \text{ V}) = -920 \text{ J/s} \quad (1)$
 $\dot{W}_{sh} = -150 \text{ J/s}$

$m = \frac{P_1 V_1}{R T_1} = \frac{(130 \text{ kPa})(3 \text{ m}^3)}{(0.287 \text{ kJ/(kg K)})(300 \text{ K})} = 4.53 \text{ kg} \quad (1)$

8

Figure 5: Insulated rigid vessel with resistance heater and paddle wheel for Q4.

Test 3 (2019)

Q1 — pure-substance diagram

Rewritten question. Label the six phase regions, critical point, triple line, constant-pressure sublimation path and constant-temperature boiling path on an unlabeled pure-substance P - v diagram.

Source page / marks. Test 3 p. 2; 5 marks.

System and boundary. Conceptual property diagram; no system boundary.

Assumptions.

- Standard pure-substance P - v topology.
- The requested boiling path passes from compressed liquid through the two-phase dome to superheated vapour.

Given / Find. Given the blank diagram; identify every phase region and requested path.

Thermodynamics A – TEST – August 2019

Question 1 – (5 marks)

Figure 1 shows an unlabelled p - v diagram for a pure substance. Fill in the following information on the diagram:

- Phase regions:
 - Liquid
 - Solid
 - Vapour
 - Liquid + Solid
 - Solid + Vapour
 - Liquid + Vapour.
- Important features:
 - Critical point
 - Triple Line.
- Process Lines: (clearly identify/label each process)
 - A constant pressure sublimation process.
 - A constant temperature boiling process. (from compressed liquid to superheated vapour)

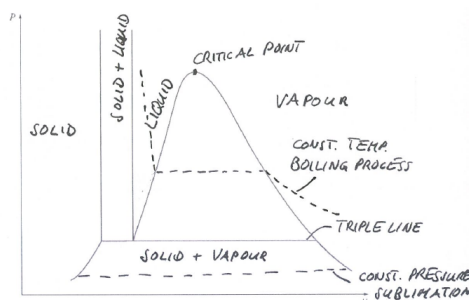


Figure 1

①
② each

Σ 5

2

Figure 6: Official labeled phase diagram for Test 3 Q1.

Clear answer. Solid is far left, liquid is between the fusion line and saturated-liquid boundary, vapour is right of the dome, liquid+vapour is inside the dome, and solid+liquid / solid+vapour occupy the corresponding coexistence regions. The critical point is at the dome apex and the triple line at the low-pressure three-phase line.

All six regions, critical point, triple line, sublimation and boiling paths correctly located.

Exam trap. The triple line and critical point are different features.

Q2 — water-steam property table

Rewritten question. Complete the missing temperature, pressure, internal energy, quality and phase entries for six water-steam states.

Source page / marks. Test 3 p. 3; 9 marks.

System and boundary. Independent equilibrium-state lookups.

Assumptions.

- Use the course saturated, superheated and compressed-water tables.
- Quality is defined only in the saturated-mixture region.

Given / Find. Use each supplied property pair to classify the phase first, then fill all missing values.

Core relations.

$$x = \frac{u - u_f}{u_{fg}}, \quad u = u_f + xu_{fg}.$$

Row (a): $x = (2000 - 610.19)/1944.2 = 0.715$. Row (d): $u = 289.24 + 0.4(2178.5) = 1160.64 \text{ kJ/kg}$.

Completed official table.

Row	$T(^{\circ}\text{C})$	$P(\text{kPa})$	$u(\text{kJ/kg})$	x / phase
a	145	415.68	2000	0.715, saturated mixture
b	50	800	209.34	–, compressed liquid
c	270	5503.0	2593.7	1.0, saturated vapour
d	69.09	30	1160.64	0.4, saturated mixture
e	500	1600	3120.1	–, superheated vapour
f	85	57.868	355.36	0.0, saturated liquid

Classify phase before choosing the table and using quality.

Exam trap. Do not force quality into rows (b) or (e).

Q3 — R-134a with heat and electrical work

Rewritten question. A piston-cylinder contains 2 kg saturated liquid R-134a and 10 kg saturated R-134a vapour at 200 kPa. At constant pressure, 250 kJ of heat is added while a 230 V source supplies a resistance element for 6 min. If the final temperature is 70°C , determine the current and show the process on a T – v diagram.

Source page / marks. Test 3 pp. 4–5; 10 marks.

System and boundary. Closed system with moving boundary; heat and electrical work enter the refrigerant.

Assumptions.

- Quasi-equilibrium constant-pressure process at 200 kPa.
- Negligible KE/PE changes.
- R-134a table properties are those used in the official solution.

Given / Find. $m_f = 2 \text{ kg}$, $m_g = 10 \text{ kg}$, $P = 200 \text{ kPa}$, $Q = 250 \text{ kJ}$, $V_e = 230 \text{ V}$, $\Delta t = 360 \text{ s}$, $T_2 = 70^{\circ}\text{C}$; find I and the process path.

Thermodynamics A – TEST – August 2019

Question 3 – (10 marks)

A piston cylinder device contains 2 kg of saturated refrigerant (R-134a) liquid and 10 kg of saturated R-134a vapour at a pressure of 200 kPa (Figure 2). Now 250 kJ of heat is transferred to the refrigerant at constant pressure while a 230 V source supplies current to a resistance element within the cylinder for a period of 6 minutes.

- Determine the current supplied if the final temperature is 70°C
- Show the process on a T-v diagram (with respect to saturation lines).

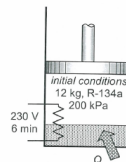


Figure 2

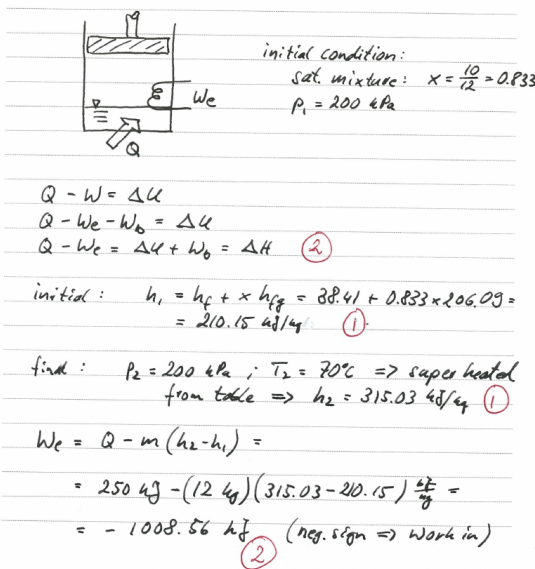


Figure 7: Piston-cylinder and official T-v path for Test 3 Q3.

Sequential working. Initial quality is the vapour mass fraction:

$$x_1 = \frac{10}{12} = 0.833$$

At 200 kPa, $h_f = 38.41$ and $h_{fg} = 206.09$ kJ/kg:

$$h_1 = h_f + x_1 h_{fg} = 38.41 + 0.833(206.09) = 210.15 \text{ kJ/kg}$$

At 200 kPa and 70°C, the final state is superheated and $h_2 = 315.03$ kJ/kg.

For a constant-pressure closed-system process, combining boundary work with ΔU gives an enthalpy balance:

$$Q + W_{e,in} = m(h_2 - h_1)$$

$$W_{e,in} = 12(315.03 - 210.15) - 250 = 1008.56 \text{ kJ}$$

$$W_{e,in} = V_e I \Delta t \Rightarrow I = \frac{1008.56 \times 10^3}{230(360)} = 12.18 \text{ A}$$

$$I = 12.18 \text{ A}$$

Exam trap. Electrical work is into the system. At constant pressure it is simplest to write the result using ΔH , which already accounts for boundary work.

Q4 — three-process ideal-gas cycle

Rewritten question. A piston-cylinder contains 0.2 kg of air initially at 1.5 MPa and 200°C. It expands at constant pressure until its volume doubles, then cools at constant volume, then is compressed polytropically back to the initial state with $n = 1.2$. Determine work for each process and net cycle work, and sketch the P-V cycle.

Source page / marks. Test 3 pp. 8–9; 13 marks.

System and boundary. Closed ideal-gas system in a piston-cylinder.

Assumptions.

- Air is ideal with $R = 0.287 \text{ kJ}/(\text{kg K})$.
- Quasi-equilibrium boundary work; negligible KE/PE changes.
- The return path obeys $PV^{1.2} = \text{constant}$.

Given / Find. $m = 0.2 \text{ kg}$, $P_1 = 1500 \text{ kPa}$, $T_1 = 473 \text{ K}$, $V_2 = 2V_1$, $V_3 = V_2$, $n = 1.2$; find W_{12} , W_{23} , W_{31} and W_{net} .

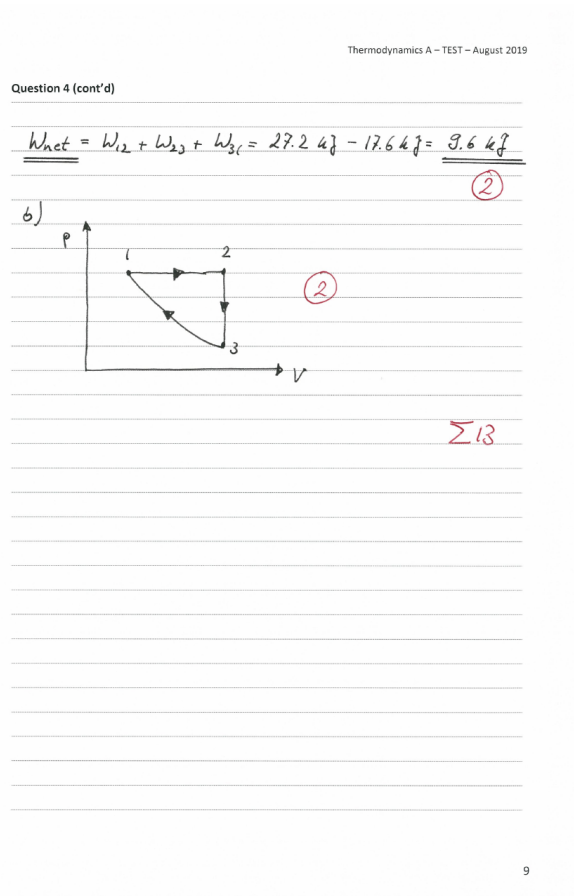


Figure 8: Official P - V cycle sketch for Test 3 Q4.

Sequential working.

$$V_1 = \frac{mRT_1}{P_1} = \frac{0.2(0.287)(473)}{1500} = 0.0181 \text{ m}^3, \quad V_2 = V_3 = 0.0362 \text{ m}^3$$

$$W_{12} = P_1(V_2 - V_1) = 1500(0.0362 - 0.0181) = 27.2 \text{ kJ}$$

$$W_{23} = 0$$

Since $P_3V_3^n = P_1V_1^n$,

$$P_3 = P_1 \left(\frac{V_1}{V_3} \right)^n = 1500(1/2)^{1.2} = 652.9 \text{ kPa}$$

$$W_{31} = \frac{P_1V_1 - P_3V_3}{1 - n} = -17.6 \text{ kJ}$$

$$W_{net} = 27.2 + 0 - 17.6 = 9.6 \text{ kJ}$$

$$\boxed{W_{12} = 27.2 \text{ kJ}, \quad W_{23} = 0, \quad W_{31} = -17.6 \text{ kJ}, \quad W_{net} = 9.6 \text{ kJ}}$$

Exam trap. Compression work is negative under the work-out convention; the net positive area means the cycle produces work.

Test 4 (2017 alternative paper)

Q1 — pure-substance P – v diagram

Rewritten question. On the unlabeled P – v diagram, identify liquid, solid, vapour, liquid+solid, solid+vapour and liquid+vapour regions; mark the critical point and triple line; draw a constant-pressure sublimation process and a constant-temperature boiling process from compressed liquid to superheated vapour.

Source page / marks. Test 4 p. 1; 5 marks.

Source note. This question repeats the same high-value diagram skill as Test 1 Q1 and Test 3 Q1.

System and boundary. Conceptual pure-substance state map; no control boundary is required.

Assumptions.

- The sketch is the standard P – v topology for a substance that contracts on freezing.
- In the liquid–vapour region, constant-temperature boiling is also constant pressure.

Given / Find. Given an unlabeled diagram; find all regions, features and the two requested paths.

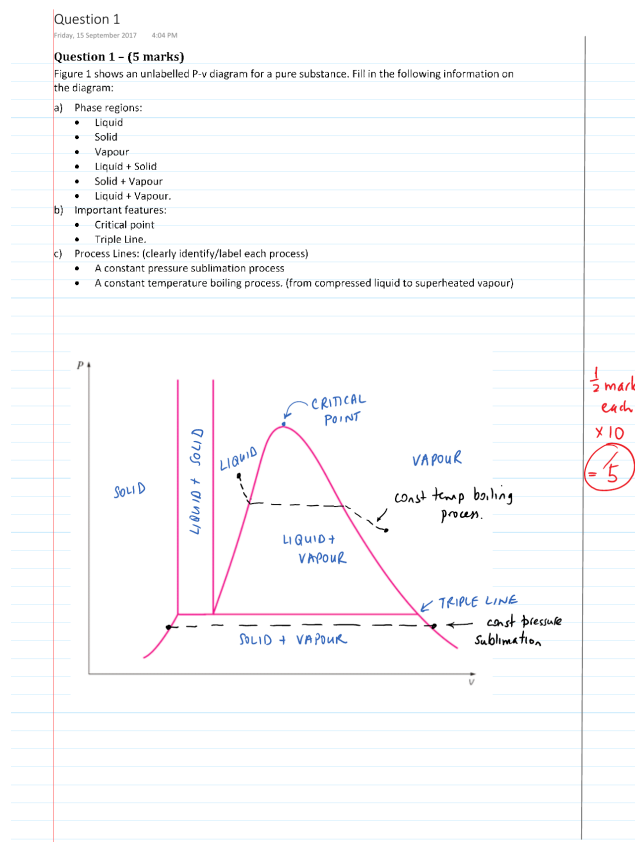


Figure 9: Fully labeled official P – v answer for Test 4 Q1.

Clear answer.

- Solid lies to the far left; liquid lies between the fusion boundary and saturated-liquid line; vapour lies right of the dome.
- Liquid+solid lies between the solid and liquid boundaries; solid+vapour lies along/below the low-pressure sublimation region; liquid+vapour lies inside the dome.
- Critical point: dome apex. Triple line: low-pressure horizontal three-phase line.
- Sublimation: horizontal constant- P path from solid to vapour below the triple point.
- Boiling: isothermal path from compressed liquid through the dome and into superheated vapour.

Label all six regions, the critical point, triple line, and both process paths as shown.

Exam trap. Do not put the triple line at the critical point.

Q2 — steam piston-cylinder with stops

Rewritten question. A piston-cylinder contains 2 kg of steam at 300 kPa with quality $x_1 = 0.25$, while its piston rests on stops. Heat is added at constant volume until 500 kPa, where the piston starts moving. Heating continues at 500 kPa until the volume is twice its initial value. Determine initial volume, boundary work, total heat transfer, and show the path on a P - v diagram.

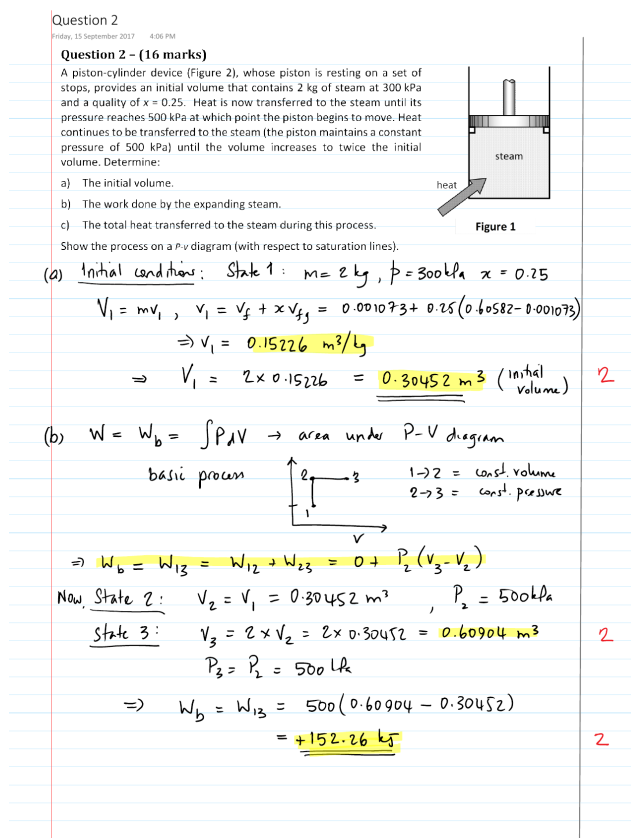
Source page / marks. Test 4 pp. 2–3; 16 marks.

System and boundary. Closed system; steam bounded by the piston-cylinder walls and moving piston.

Assumptions.

- Quasi-equilibrium piston motion; negligible kinetic and potential energy changes.
- $1 \rightarrow 2$ is constant volume and $2 \rightarrow 3$ is constant pressure.
- Saturated-water tables apply at states 1 and 3.

Given / Find. $m = 2$ kg, $P_1 = 300$ kPa, $x_1 = 0.25$, $P_2 = P_3 = 500$ kPa and $V_3 = 2V_1$; find V_1 , W , Q , and process path.



Mid-sem test (ALT) Page 2

Figure 10: Piston-cylinder apparatus and official P - v path for Test 4 Q2.

Property values.

$$300 \text{ kPa: } v_f = 0.001073, \quad v_g = 0.60582 \text{ m}^3/\text{kg}, \quad u_f = 561.11, \quad u_{fg} = 1982.1 \text{ kJ/kg},$$

$$500 \text{ kPa: } v_f = 0.001093, \quad v_g = 0.37483 \text{ m}^3/\text{kg}, \quad u_f = 639.54, \quad u_{fg} = 1921.2 \text{ kJ/kg}.$$

Sequential working.

$$v_1 = v_f + x_1(v_g - v_f) = 0.001073 + 0.25(0.60582 - 0.001073) = 0.15226 \text{ m}^3/\text{kg}$$

$$V_1 = mv_1 = 2(0.15226) = 0.30452 \text{ m}^3, \quad V_3 = 2V_1 = 0.60904 \text{ m}^3$$

$$W_{12} = 0 + P_2(V_3 - V_2) = 500(0.60904 - 0.30452) = 152.26 \text{ kJ}$$

$$u_1 = 561.11 + 0.25(1982.1) = 1056.64 \text{ kJ/kg}$$

$$v_3 = V_3/m = 0.30452 \text{ m}^3/\text{kg}, \quad x_3 = \frac{0.30452 - 0.001093}{0.37483 - 0.001093} = 0.812$$

$$u_3 = 639.54 + 0.812(1921.2) = 2199.31 \text{ kJ/kg}$$

$$Q_{13} = W_{13} + m(u_3 - u_1) = 152.26 + 2(2199.31 - 1056.64) = 2437.6 \text{ kJ}$$

$$V_1 = 0.30452 \text{ m}^3, \quad W_{out} = 152.26 \text{ kJ}, \quad Q_{in} = 2437.6 \text{ kJ}$$

Exam trap. Use total volume in $P\Delta V$, but specific volume in table-quality calculations.

Q3 — heating a room

Rewritten question. A $5 \times 6 \times 3 \text{ m}$ room at atmospheric pressure is heated by a steam radiator supplying $15,000 \text{ kJ/h}$. A 200 W fan distributes the air, and heat loss is 100 kJ/min . The room is heated from 7°C to 22°C . (a) Estimate the heating time by treating room air as a constant-volume system. (b) Discuss whether that assumption is reasonable.

Source page / marks. Test 4 pp. 4–5; 10 marks.

System and boundary. Approximate closed, rigid system containing the room air.

Assumptions.

- Ideal-gas air; atmospheric pressure 101.3 kPa initially.
- Uniform room-air temperature; negligible KE/PE changes.
- Radiator input, fan work and stated heat loss are constant during heating.

Given / Find. $V = 90 \text{ m}^3$, $T_1 = 280 \text{ K}$, $T_2 = 295 \text{ K}$ and the three energy rates; find Δt and judge the model.

Question 3
Index: 10 September 2017 4:13 PM

Question 3 - (10 marks)

A $5 \text{ m} \times 6 \text{ m} \times 3 \text{ m}$ room at atmospheric pressure is heated by the radiator of a steam heating system (Figure 3). The steam radiator transfers heat at a rate of $15,000 \text{ kJ/hour}$ and a 200 W fan is used to distribute the warm air around the room. The rate of heat loss from the room is estimated to be around 100 kJ/min . The initial temperature of the room is 7°C , and it is desired to heat the room to 22°C .

a) Determine how long it will take to heat the air in the room assuming the room is a constant volume system. Also, state any other assumptions you make.

b) Is it reasonable to assume that the room is a constant volume system? Justify your answer.

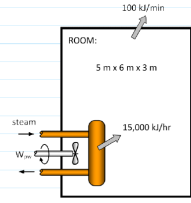


Figure 1

(a) First Law - closed system (energy balance)

$$Q - W = \Delta U = m(u_2 - u_1) = m C_v \Delta T$$

$$\Rightarrow (\dot{Q} - \dot{W}) \Delta t = m(u_2 - u_1) = m C_v \Delta T$$

Need mass: ideal gas

initial conditions: $p_1 = \text{atmospheric} = 101.3 \text{ kPa}$
 $V_1 = 5 \times 6 \times 3 = 90 \text{ m}^3$
 $T_1 = 7^\circ\text{C} = 280 \text{ K}$

$$\Rightarrow m = \frac{p_1 V_1}{R T_1} = \frac{101.3 \times 90}{0.287 \times 280} = 113.45 \text{ kg}$$

If we use Table A-17: $u_1 @ 280 \text{ K} = 199.75 \text{ kJ/kg}$

final state: $T_2 = 22^\circ\text{C} = 295 \text{ K} \Rightarrow u_2 = 210.49 \text{ kJ/kg}$

$$\Rightarrow (\dot{Q} - \dot{W}) \Delta t = m(u_2 - u_1)$$

$$\Rightarrow \left[\frac{15,000 \text{ kJ}}{3600 \text{ s}} - \frac{100 \text{ kJ}}{60 \text{ s}} - \left(-\frac{200 \text{ W}}{1000 \text{ W}} \cdot \frac{\text{kJ}}{\text{s}} \right) \right] \Delta t = 113.45 (210.49 - 199.75)$$

Mid-sem test (ALT) Page 4

Figure 11: Room, radiator, fan and heat-loss diagram for Test 4 Q3.

Sequential working.

$$m = \frac{PV}{RT_1} = \frac{101.3(90)}{0.287(280)} = 113.45 \text{ kg}$$

From ideal-gas Table A-17, $u_1 = 199.75$ at 280 K and $u_2 = 210.49 \text{ kJ/kg}$ at 295 K :

$$\Delta U = m(u_2 - u_1) = 113.45(210.49 - 199.75) = 1218.47 \text{ kJ}$$

Net input rate:

$$\dot{E}_{in} = \frac{15000}{3600} + 0.200 - \frac{100}{60} = 2.70 \text{ kJ/s}$$

$$\Delta t = \frac{1218.47}{2.70} = 451.3 \text{ s} = 7.52 \text{ min}$$

Using average $c_v = 0.7175 \text{ kJ/(kg K)}$ gives 7.54 min.

$$\Delta t \approx 7.52\text{--}7.54 \text{ min}$$

Assumption discussion. The geometric room volume is fixed, but a real room is not an airtight closed system; some air escapes as it warms. Constant volume is nevertheless a reasonable exam approximation because the temperature rise and associated leakage are modest. Constant pressure would require a mass-flow model.

Exam trap. Fan work enters the air, while the stated 100 kJ/min leaves it.

Q4 — R-134a property table

Rewritten question. Complete the missing T , P , h , quality and phase entries for six R-134a states.

Source page / marks. Test 4 p. 6; 9 marks.

System and boundary. Independent equilibrium-state lookups; no process boundary.

Assumptions.

- Use the course R-134a saturation, superheated-vapour and compressed-liquid tables.
- Quality applies only in the two-phase region.

Given / Find. Use each supplied property pair to identify phase and fill all missing entries.

Working pattern.

$$x = \frac{h - h_f}{h_{fg}} \quad \text{for a mixture,} \quad h \approx h_f(T) \quad \text{for a compressed liquid.}$$

For row (a), $x = (250 - 102.34)/167.19 = 0.883$. For row (c), $h = 30.67 + 0.4(210.23) = 114.762 \text{ kJ/kg}$.

Completed official table.

Row	$T(^{\circ}\text{C})$	$P(\text{kPa})$	$h(\text{kJ/kg})$	x / phase
a	36	912.35	250	0.88, saturated mixture
b	50	500	291.98	–, superheated vapour
c	–16	157.38	114.762	0.4, saturated mixture
d	8.91	400	255.61	1.0, saturated vapour
e	–32	76.71	10.09	0.0, saturated liquid
f	–2	900	49.15	–, compressed liquid

Use phase first, then select the saturation, superheat, or compressed-liquid table.

Exam trap. Do not assign a quality to superheated or compressed-liquid states.

Test 2025

Q1

Rewritten question. A 10 kg system was taken from 25 °C to 75 °C. Given the specific heat capacity of the system is $c_p = 3000 \text{ J}/(\text{kg} \cdot ^\circ \text{C})$.¹

Source page / marks. Question sheet p. 1; Q1 worth 9 marks, split as (a) 4 marks, (b) 3 marks, (c) 2 marks.

System and boundary. Closed system; no mass crosses the boundary. The energy accounting is for the system only.

Assumptions.

- The specific heat is constant over the temperature change.
- No work interaction is stated, so the heating is interpreted as pure sensible energy addition.
- Heat losses are only introduced explicitly in part (c).

Given / Find.

- Given: $m = 10 \text{ kg}$, $T_1 = 25 ^\circ \text{C}$, $T_2 = 75 ^\circ \text{C}$, $c_p = 3000 \text{ J}/(\text{kg} \cdot ^\circ \text{C})$.
- Find: (a) Q added, (b) $\dot{Q}$, (c) heat lost relative to the no-loss case.

Governing model.

$$Q = mc_p \Delta T, \quad \dot{Q} = \frac{Q}{\Delta t}$$

Sequential working with units.

$$\Delta T = T_2 - T_1 = 75 - 25 = 50 ^\circ \text{C}$$

$$Q = (10 \text{ kg})(3000 \text{ J}/(\text{kg} \cdot ^\circ \text{C}))(50 ^\circ \text{C}) = 1.50 \times 10^6 \text{ J} = 1500 \text{ kJ}$$

$$\dot{Q} = \frac{1500 \text{ kJ}}{20 \text{ s}} = 75 \text{ kW}$$

$$Q_{\text{ideal}} = (10)(3000)(80 - 25) = 1.65 \times 10^6 \text{ J} = 1650 \text{ kJ}$$

$$Q_{\text{lost}} = Q_{\text{ideal}} - Q_{\text{actual}} = 1650 - 1500 = 150 \text{ kJ}$$

Boxed answers.

$Q = 1500 \text{ kJ}, \quad \dot{Q} = 75 \text{ kW}, \quad Q_{\text{lost}} = 150 \text{ kJ}$
--

Exam check. Units reduce correctly: $\text{J}/(\text{kg} \cdot ^\circ \text{C}) \times \text{kg} \times ^\circ \text{C} = \text{J}$. The loss in part (c) is the difference between the ideal no-loss heating requirement and the actual heating supplied.

Q2

Rewritten question. 2 kg of water initially at 95 °C with quality 0.35 occupies a linear spring-loaded piston–cylinder. It is heated until the pressure reaches 800 kPa and the temperature is 275 °C. Determine the initial and final specific volumes and total work.

Source page / marks. Question sheet p. 1; Q2 worth 12 marks, split as (a) 3 marks, (b) 4 marks, (c) 5 marks.

System and boundary. Closed system; piston–cylinder with a moving boundary. The boundary does work only while the piston moves.

Assumptions.

- The piston motion is quasi-equilibrium and the spring gives a linear P – V path.
- The initial state is a saturated mixture at 95 °C, so saturation data apply.
- The final state is superheated at 800 kPa and 275 °C, so superheated-steam data apply.
- Kinetic and potential energy changes are negligible.

Given / Find.

- Given: $m = 2 \text{ kg}$, $T_1 = 95 ^\circ \text{C}$, $x_1 = 0.35$, $P_2 = 800 \text{ kPa}$, $T_2 = 275 ^\circ \text{C}$.
- Find: v_1 , v_2 , and $W_{1 \rightarrow 2}$.

Governing model.

$$v_1 = v_f + x_1(v_g - v_f), \quad v_2 = v(P_2, T_2), \quad W_{1 \rightarrow 2} = \int_1^2 P dV$$

For a linear spring path,

$$W_{1 \rightarrow 2} = \frac{P_1 + P_2}{2}(V_2 - V_1) = m \frac{P_1 + P_2}{2}(v_2 - v_1)$$

¹Source sheet: p. 1, 9 marks total.

Source table values. Using local steam tables for water:

$$\begin{aligned} P_{\text{sat}}(95^\circ\text{C}) &\approx 84.6 \text{ kPa}, \\ v_f(95^\circ\text{C}) &\approx 0.00104 \text{ m}^3/\text{kg}, \\ v_g(95^\circ\text{C}) &\approx 2.01 \text{ m}^3/\text{kg}, \\ \text{and at } 800 \text{ kPa, } 275^\circ\text{C} : v_2 &\approx 0.307 \text{ m}^3/\text{kg} \text{ (superheated-table lookup; approximate)}. \end{aligned}$$

If a table entry is read as a symbol rather than a fully trusted number, the calculation below is shown with a clearly labelled approximation.

Sequential working with units.

$$v_1 = 0.00104 + 0.35(2.01 - 0.00104) \approx 0.704 \text{ m}^3/\text{kg}$$

$$V_1 = mv_1 = (2 \text{ kg})(0.704 \text{ m}^3/\text{kg}) = 1.408 \text{ m}^3$$

$$V_2 = mv_2 \approx (2 \text{ kg})(0.307 \text{ m}^3/\text{kg}) = 0.614 \text{ m}^3$$

Because the device is spring-loaded, the pressure varies linearly between states, so using the average pressure gives the same boundary work as the area under the straight line.

$$W_{1 \rightarrow 2} \approx \frac{84.6 + 800}{2} \text{ kPa} (0.614 - 1.408) \text{ m}^3$$

$$W_{1 \rightarrow 2} \approx (442.3 \text{ kPa})(-0.794 \text{ m}^3) \approx -351 \text{ kJ}$$

Boxed answers.

$v_1 \approx 0.704 \text{ m}^3/\text{kg},$	$v_2 \approx 0.307 \text{ m}^3/\text{kg},$	$W_{1 \rightarrow 2} \approx -351 \text{ kJ}$
--	--	---

Exam check and source warning. The computed path is a net compression, so work by the water is negative even though heat is supplied. This conflicts with the prompt's phrase "work produced" and suggests that at least one printed state value may be erroneous. Preserve the literal calculation above, but confirm the intended final state with the lecturer before using it as a model answer.

Q3

Rewritten question. R-134a at 0.8 MPa and 90 °C is cooled at 0.8 MPa to 30 °C by air. Air enters at 100 kPa and 22 °C with a volume flow rate of 400 m³/min and leaves at 95 kPa and 57 °C. Determine the inlet/outlet enthalpies of both streams and the refrigerant mass flow rate.

Source page / marks. Question sheet p. 2; Q3 worth 9 marks, split as (a) 2 marks, (b) 2 marks, (c) 1 mark, (d) 1 mark, (e) 3 marks.

System and boundary. Two coupled steady-flow control volumes: one for the R-134a stream and one for the air stream. The condenser wall is the interaction surface between them.

Assumptions.

- Steady flow through both streams.
- No shaft work and negligible kinetic/potential energy changes.
- Air is an ideal gas, so $h = h(T)$ only.
- The condenser is adiabatic to the surroundings, so heat lost by the refrigerant equals heat gained by the air.

Given / Find.

- Given: R-134a at $P_3 = 0.8 \text{ MPa}$, $T_3 = 90^\circ\text{C}$, $P_4 = 0.8 \text{ MPa}$, $T_4 = 30^\circ\text{C}$; air at $P_1 = 100 \text{ kPa}$, $T_1 = 22^\circ\text{C}$, $\dot{V}_1 = 400 \text{ m}^3/\text{min}$, $P_2 = 95 \text{ kPa}$, $T_2 = 57^\circ\text{C}$.
- Find: h_1, h_2, h_3, h_4 , and $\dot{m}_{\text{R134a}}$.

Governing model.

$$\dot{Q}_{\text{air}} = \dot{m}_{\text{air}}(h_4 - h_3), \quad \dot{Q}_{\text{R}} = \dot{m}_{\text{R134a}}(h_2 - h_1), \quad \dot{Q}_{\text{air}} + \dot{Q}_{\text{R}} = 0$$

with the air mass flow rate from the ideal-gas relation,

$$\dot{m}_{\text{air}} = \rho_1 \dot{V}_1 = \frac{P_1}{RT_1} \dot{V}_1$$

Source table values. From the local property tables / worked resource:

$$\begin{aligned} h_1(\text{R134a, } 0.8 \text{ MPa, } 90^\circ\text{C}) &\approx 416.03 \text{ kJ/kg}, \\ h_2(\text{R134a, } 0.8 \text{ MPa, } 30^\circ\text{C}) &\approx 218.19 \text{ kJ/kg}, \\ h_3(\text{air, } 22^\circ\text{C}) &\approx 295.0 \text{ kJ/kg} \text{ (Table A-17; approximate)}, \\ h_4(\text{air, } 57^\circ\text{C}) &\approx 330.8 \text{ kJ/kg} \text{ (Table A-17; approximate)}. \end{aligned}$$

Hence,

$$h_1 - h_2 \approx 197.84 \text{ kJ/kg}, \quad h_4 - h_3 \approx 35.8 \text{ kJ/kg}$$

Sequential working with units.

$$\dot{m}_{\text{air}} = \frac{(100 \text{ kPa})(400 \text{ m}^3/\text{min})}{(0.2870 \text{ kJ}/(\text{kg K}))(295.15 \text{ K})} \times \frac{1 \text{ min}}{60 \text{ s}} \approx 7.92 \text{ kg/s}$$

$$\dot{Q}_{\text{air}} = \dot{m}_{\text{air}}(h_4 - h_3) \approx (7.92 \text{ kg/s})(35.8 \text{ kJ/kg}) \approx 284 \text{ kW}$$

For the refrigerant side, the heat rejected has the same magnitude:

$$\dot{Q}_{\text{R}} = \dot{m}_{\text{R134a}}(h_2 - h_1)$$

$$\dot{m}_{\text{R134a}} = \frac{\dot{Q}_{\text{air}}}{h_1 - h_2} \approx \frac{284 \text{ kW}}{197.84 \text{ kJ/kg}} \approx 1.44 \text{ kg/s}$$

Boxed answers.

$h_1 \approx 416.03 \text{ kJ/kg}, \quad h_2 \approx 218.19 \text{ kJ/kg}, \quad h_3 \approx 295.0 \text{ kJ/kg}, \quad h_4 \approx 330.8 \text{ kJ/kg}, \quad \dot{m}_{\text{R134a}} \approx 1.44 \text{ kg/s}$
--

Exam check. The air and refrigerant balances must oppose each other in sign. If the air enthalpy rise is computed from local table values instead of the rounded approximations above, the refrigerant mass flow rate should move slightly but remain of the same order; do not report false precision beyond the table accuracy.

Selected end-of-year questions

Scope note. These are stretch applications of the Week 5 steady-flow balance and Week 6 refrigerator/COP ideas. They also require later vapour-compression-cycle conventions and R-134a property tables. Otto, Diesel, Brayton-regenerator and Rankine questions were excluded because their cycle theory lies beyond Weeks 1–6.

End-of-year Problem 10 — commercial refrigerator

Rewritten question. A commercial refrigerator using R-134a keeps a refrigerated space at -25°C . Cooling water enters the condenser at 20°C at 0.25 kg/s and leaves at 25°C . Refrigerant enters the condenser at 1.2 MPa and 70°C and leaves at 42°C . It enters the compressor at 60 kPa and -20°C . The compressor gains 450 W of heat from the surroundings. Determine (a) quality at the evaporator inlet, (b) refrigeration load and (c) refrigerator COP.

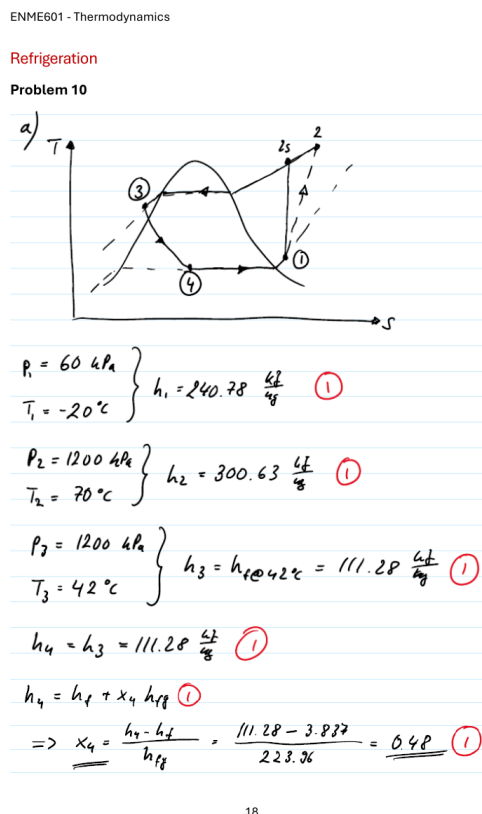
Source page / marks. Previous Test and Exam Problems, p. 6, Problem 10; official solution pp. 18–19.

System and boundary. Steady-flow control volumes around the condenser, compressor, throttle and evaporator.

Assumptions.

- Steady operation; negligible kinetic and potential energy changes.
- No pressure loss through each heat exchanger.
- Throttling is isenthalpic: $h_4 = h_3$.
- Liquid-water enthalpy is taken from the water table; R-134a properties are taken from the course tables.

Given / Find. Given the four refrigerant state conditions, cooling-water data and compressor heat gain; find x_4 , $\dot{Q}_L$ and COP_R .



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Figure 12: Official T - s sketch and state numbering for end-of-year Problem 10.

Property values.

$$h_1 = 240.78, \quad h_2 = 300.63, \quad h_3 = h_4 = 111.28 \quad \text{kJ/kg}$$

At 60 kPa , $h_f = 38.87$ and $h_{fg} = 223.36\text{ kJ/kg}$. For the cooling water, $h_{w,in} = 83.915$ and $h_{w,out} = 104.83\text{ kJ/kg}$.

Sequential working.

$$x_4 = \frac{h_4 - h_f}{h_{fg}} = \frac{111.28 - 38.87}{223.36} = 0.324 \approx 0.32$$

Condenser energy balance:

$$\begin{aligned}\dot{m}_w(h_{w,out} - h_{w,in}) &= \dot{m}_R(h_2 - h_3) \\ \dot{m}_R &= \frac{0.25(104.83 - 83.915)}{300.63 - 111.28} = 0.0276 \approx 0.028 \text{ kg/s} \\ \dot{Q}_H &= \dot{m}_R(h_2 - h_3) = 0.028(300.63 - 111.28) = 5.30 \text{ kW}\end{aligned}$$

For the compressor, heat enters at 0.45 kW:

$$\dot{W}_{in} = \dot{m}_R(h_2 - h_1) - \dot{Q}_{comp,in} = 0.028(300.63 - 240.78) - 0.45 = 1.23 \text{ kW}$$

For the complete cycle, $\dot{Q}_H = \dot{Q}_L + \dot{W}_{in} + \dot{Q}_{comp,in}$:

$$\dot{Q}_L = 5.30 - 1.23 - 0.45 = 3.62 \text{ kW}$$

$$\text{COP}_R = \frac{\dot{Q}_L}{\dot{W}_{in}} = \frac{3.62}{1.23} = 2.94$$

Boxed official answers.

$x_4 \approx 0.32, \quad \dot{m}_R \approx 0.028 \text{ kg/s}, \quad \dot{Q}_L = 3.62 \text{ kW}, \quad \dot{W}_{in} = 1.23 \text{ kW}, \quad \text{COP}_R = 2.94$
--

Exam trap. The condenser-water heat rate is $\dot{Q}_H$, not $\dot{Q}_L$, and the compressor is not adiabatic.

End-of-year Problem 11 — evaporator pressure and a 250 kW cycle

Rewritten question. (A) Briefly explain the main factors influencing evaporator-pressure choice in a vapour-compression refrigerator. (B) A 250 kW R-134a refrigerator receives saturated vapour at 140 kPa at the compressor inlet and compresses it to 800 kPa with compressor isentropic efficiency 85%. Refrigerant is saturated liquid before throttling. Sketch the cycle, determine quality after throttling, compressor input power, and COP.

Source page / marks. Previous Test and Exam Problems, pp. 6–7, Problem 11; official solution p. 20.

System and boundary. Four steady-flow components forming a simple vapour-compression cycle.

Assumptions.

- Steady operation and negligible kinetic/potential energy changes.
- Compressor inlet is saturated vapour and throttle inlet is saturated liquid exactly as stated.
- Throttle is isenthalpic; compressor is adiabatic with $\eta_c = 0.85$.

ENME601 - Thermodynamics

Problem 11**Part A:**

- Must evaporate at a pressure about 5 to 10°C below that of fridge space.
- Pressure must not be below atmospheric; will cause air leakage into system.

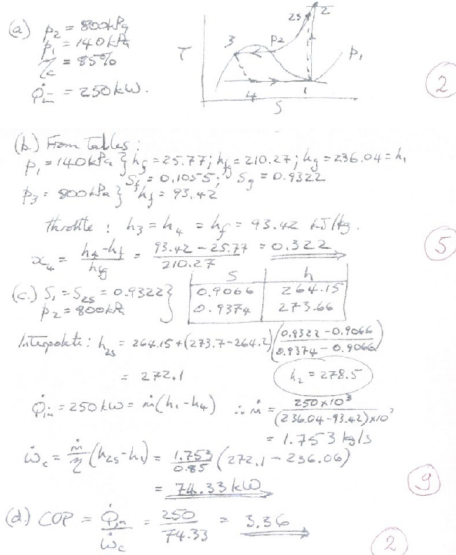
Part B:

Figure 13: Official state sketch and property working for end-of-year Problem 11.

Part A — clear conceptual answer.

- The refrigerant saturation temperature should normally be about 5–10°C below the refrigerated-space temperature so heat has a practical driving temperature difference.
- Evaporator pressure should not be below atmospheric pressure where avoidable, because air leakage into the system becomes likely.
- A very low evaporator pressure increases compressor pressure ratio, work and discharge temperature, reducing efficiency and reliability.

Property values from the official solution.

$$h_1 = 236.04, \quad s_1 = 0.9322, \quad h_3 = h_4 = 95.42 \quad \text{kJ/kg}$$

At 800 kPa and $s_{2s} = s_1$, interpolation gives $h_{2s} = 267.10 \text{ kJ/kg}$.

Sequential working. After throttling at the 140 kPa evaporator pressure,

$$x_4 = \frac{h_4 - h_f}{h_{fg}} = \frac{95.42 - 25.77}{210.27} = 0.332$$

Compressor efficiency:

$$\eta_c = \frac{h_{2s} - h_1}{h_2 - h_1} \Rightarrow h_2 = h_1 + \frac{h_{2s} - h_1}{0.85} = 272.1 \text{ kJ/kg}$$

Evaporator load determines refrigerant flow:

$$250 = \dot{m}_R(h_1 - h_4) \Rightarrow \dot{m}_R = \frac{250}{236.04 - 95.42} = 1.78 \text{ kg/s}$$

$$\dot{W}_{in} = \dot{m}_R(h_2 - h_1) = 1.78(272.1 - 236.04) = 64.3 \text{ kW}$$

The official handwritten page reports 74.33 kW and hence $\text{COP} = 3.36$; its intermediate mass-flow/enthalpy arithmetic is internally inconsistent with the values printed on that same page. Using the printed properties consistently gives:

$$\text{COP}_R = \frac{250}{64.3} = 3.89$$

Boxed answer with source discrepancy.

$$x_4 \approx 0.332, \quad \dot{m}_R \approx 1.78 \text{ kg/s}, \quad \dot{W}_{in} \approx 64.3 \text{ kW}, \quad \text{COP}_R \approx 3.89$$

Official-page note: the handwritten sheet boxes $\dot{W}_{in} = 74.33 \text{ kW}$ and $\text{COP} = 3.36$. These do not follow from its displayed h_1, h_{2s}, h_3 , 250 kW load and 85% efficiency. The internally consistent reconstruction above is recommended; retain the source values for lecturer comparison.

Exam trap. Do not apply compressor efficiency directly to temperature, and do not hide inconsistent source arithmetic.

End-of-year Problem 12 — condenser heat exchanger

Rewritten question. An R-134a vapour-compression refrigerator maintains a space at -30°C . Refrigerant enters the condenser at 1.2 MPa and 65°C and leaves at 42°C . It enters the compressor at 60 kPa and -34°C , while the compressor gains 450 W from the surroundings. Cooling water enters the condenser at 0.25 kg/s and 18°C and leaves at 26°C . Sketch the cycle; determine refrigeration capacity, compressor input and COP; explain why the compressor inlet is superheated.

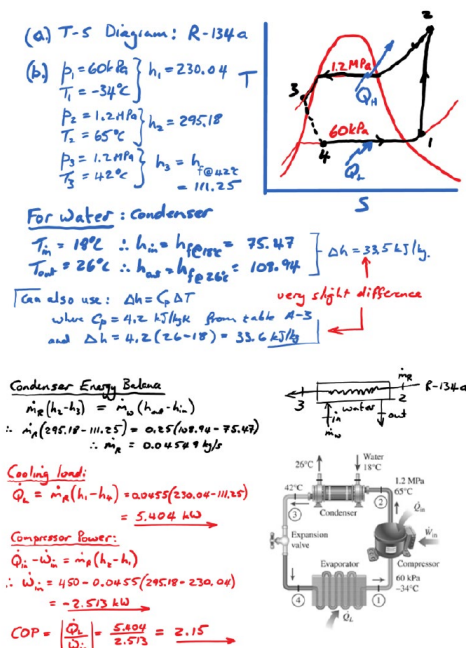
Source page / marks. Previous Test and Exam Problems, pp. 7–8, Problem 12; official solution pp. 21–22.

System and boundary. Refrigeration loop plus a separate condenser-water control volume.

Assumptions.

- Steady operation; negligible kinetic/potential energy changes and heat loss from the condenser to surroundings.
- Condenser and evaporator pressures are 1.2 MPa and 60 kPa.
- Throttling is isenthalpic.

ENME601 - Thermodynamics

Problem 12

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Figure 14: Official cycle, condenser diagram and T-s sketch for end-of-year Problem 12.

Official property values.

$$h_1 = 230.04, \quad h_2 = 295.18, \quad h_3 = h_4 = 111.25 \quad \text{kJ/kg}$$

For water, $h_{w,in} = 75.47$ and $h_{w,out} = 109.94 \text{ kJ/kg}$, so $\Delta h_w = 33.57 \text{ kJ/kg}$.

Sequential working. Condenser balance:

$$\dot{m}_R(h_2 - h_3) = \dot{m}_w(h_{w,out} - h_{w,in})$$

$$\dot{m}_R = \frac{0.25(109.04 - 75.47)}{295.18 - 111.25} = 0.0456 \text{ kg/s}$$

$$\dot{Q}_L = \dot{m}_R(h_1 - h_4) = 0.0456(230.04 - 111.25) = 5.40 \text{ kW}$$

With 0.45 kW entering the compressor,

$$\dot{W}_{in} = \dot{m}_R(h_2 - h_1) - 0.45 = 0.0456(295.18 - 230.04) - 0.45 = 2.52 \text{ kW}$$

Using the source's rounded mass flow gives 2.575 kW and

$$\text{COP}_R = \frac{5.404}{2.575} = 2.10 \text{ (the handwritten page labels this approximately 2.15).}$$

Boxed answers.

$\dot{m}_R \approx 0.0456 \text{ kg/s}, \quad \dot{Q}_L \approx 5.40 \text{ kW}, \quad \dot{W}_{in} \approx 2.52\text{--}2.58 \text{ kW}, \quad \text{COP}_R \approx 2.10\text{--}2.15$

Why superheat the compressor inlet? A compressor is designed for vapour, not a liquid–vapour mixture. Liquid droplets can damage valves, pistons or blades. Slight superheat ensures dry vapour enters the compressor.

Exam trap. Do not confuse condenser effectiveness with COP, and include the compressor heat gain with the correct sign.

2 Final exam-use checklist

1. Draw the boundary and name the system type before selecting an equation.
2. Build a state table and classify each phase before reading properties.
3. Write the full energy balance before deleting zero terms.
4. Separate total and specific quantities; PV is energy only with compatible units.
5. Mark constant- V , constant- P , linear-spring and polytropic paths on the diagram.
6. For electrical devices, determine whether $VI\Delta t$ is work into or out of the system.
7. For a heat exchanger, heat lost by one stream equals heat gained by the other if external losses are neglected.
8. For refrigerators, keep $\dot{Q}_H$, $\dot{Q}_L$ and $\dot{W}_{in}$ distinct and state the COP definition.
9. Box a value with units and perform one physical direction/sign check.
10. If the provided source data conflict, state the conflict instead of silently changing them.